



LEARNING MANAGEMENT SYSTEM

Version 1 Product Brief

A focused, beta-phase platform delivering e-learning, virtual classes, and academic content to senior secondary school students in Kenya — starting with holiday support.

PRODUCT OWNER

Ms Valary Lambisia

DEVELOPMENT PARTNER

Zafrika Tech Lab

STATUS

Beta — Proof of Concept

01 · OVERVIEW

What We Are Building

Hone Your Mark is a Kenya-focused Learning Management System targeting Form 1–4 students in senior secondary schools. Version 1 is a lean, beta release designed to validate the core concept: connecting students to quality academic content and live virtual classes during holiday periods.

This document outlines every feature planned for Version 1, prepared for review and sign-off by the Founder. It is intended to align expectations between Hone Your Mark and Zafrika Tech Lab before development begins.



Target Users

Form 1–4 students and their teachers in Kenyan senior secondary schools.
Students aged approximately 13–18 years.



Primary Use Window

School holidays — when students lack structured learning. The system must be operational and compelling during these windows.



Core Hypothesis

Students and parents will pay for curated academic content and teacher-led virtual classes delivered through a simple, reliable platform.



Business Model (V1)

Fee-gated e-library access and e-classes, collected by teachers/admins and manually enabled per learner.

01 • OBJECTIVES

Goals for Version 1

The following objectives guide every feature decision in this document. Any feature that does not serve at least one of these objectives is deferred to a later version.

- 1 Demonstrate that an LMS can work for the Kenyan secondary school context, validating willingness to pay.
- 2 Give teachers a simple toolkit to publish content, run virtual classes, assign work, and track student progress.
- 3 Give students safe, structured access to content and classes without unnecessary complexity.
- 4 Keep onboarding fast — a student or teacher should be active in under 5 minutes.
- 5 Collect real user feedback during the first holiday window to inform Version 2.



Beta Boundary

Version 1 is intentionally basic. Complex features (AI tutoring, deep analytics, third-party integrations beyond Google Meet, automated payment gateways) are explicitly out of

scope and will be planned for V2 based on learnings.

01 · SCOPE

Scope & Constraints

AREA	IN SCOPE (V1)	OUT OF SCOPE
Payments	Manual confirmation by Teacher-Admin	M-Pesa or card gateway automation
Content types	Text, images, embedded videos (YouTube/uploaded)	PDF upload by teachers, SCORM
Video hosting	Upload to Youtube, store in e-library	Live streaming on platform
Virtual classes	Google Meet scheduling & link sharing	In-platform video conferencing
Student output	Students can download library content as PDF	Collaborative documents, wikis
Notifications	In-app	Push notifications (mobile app)
Platform	Web (responsive, mobile-friendly browser)	Native iOS/Android app

02 · ONBOARDING

User Registration

Both user types register through a guided multi-step form. All accounts are subject to approval or admin activation before full access is granted.



Student Registration

With parent/guardian consent

REQUIRED BIO DATA

- | | |
|------------------------------|----------------------------|
| – Full name | – Date of birth |
| – Gender | – School name |
| – Current form / class level | – County / sub-county |
| – Subjects enrolled | – Profile photo (optional) |

PARENT / GUARDIAN DETAILS (CONSENT)

- | | |
|-----------------------------|-------------------------------|
| – Parent/guardian full name | – Relationship to student |
| – Phone number | – Email address |
| – National ID number | – Consent checkbox (required) |

SYSTEM-GENERATED CREDENTIALS

Upon registration approval, the system auto-generates:

- ➔ **Admission Number** — a unique alphanumeric identifier (e.g., HYM-2025-00143). This is the student's login username.
- ➔ **Default Password** — the student's date of birth in DDMMYYYY format (e.g., 01012010 for 1 January 2010).



Admission Number Delivery

The admission number and temporary password are displayed on-screen after registration is approved.



Teacher Registration

Manually approved by platform admin

REQUIRED BIO DATA

— Full name

— Date of birth

— Gender

— National ID / TSC number

— Phone number

— Email address

— County / location

— Profile photo

PROFESSIONAL DETAILS

— Subjects taught (multi-select)

— Forms / levels covered

— Years of experience

— Current school affiliation (optional)

Teachers may also be elevated to **Teacher-Admin** role by the platform super-admin, granting access to student management and payment confirmation tools.

03 · TEACHER FEATURES

What Teachers Can Do

Teachers are the core content and instruction providers on the platform. Their feature set is designed to be simple yet powerful enough to deliver real academic value.

Academic Content Creation

Teachers create structured content that appears in the e-library and is available for enrolled students.

1 Rich text content — A simple editor supporting headings, paragraphs, bold/italic, and numbered/bulleted lists.

2 Image embedding — Upload images (JPG, PNG) directly into content. Images are stored on the platform.

3 Video content — Record or upload a teaching video; videos are stored under the teacher's Hone Your Mark account and added to the e-library. (Note: No PDF uploads in V1.)

4 Content organisation — Attach a subject, form level, and topic tag to every piece of content for easy filtering.

- 5 **Draft / Publish** — Content can be saved as a draft before being published to the e-library.



PDF Download by Students

Although teachers cannot upload PDFs, published content (text + images) can be downloaded by students as a system-generated PDF — preserving formatting and images.

Assignments & Assessment

- 1 **Create assignment** — Write an assignment using the text editor. Attach a due date, subject, and form level. Assign to a specific student, a group, or all enrolled students.
- 2 **View submissions** — A simple list view of which students have submitted, with submission date and content.
- 3 **Mark & feedback** — Enter a numeric score and written feedback per submission. Marked assignments are visible to the respective student.

Virtual Class Scheduling

- 1 **Create class event** — Enter class title, subject, form, date, time, and duration. The system generates a Google Meet link automatically via the Google Calendar/Meet API.
- 2 **Share class link** — The Meet link is displayed on the student's class schedule and sent via in-app notification.
- 3 **Class visibility** — Unpaid students see upcoming classes in an *inactive/preview* state — they can see the title, date, and subject but not the Join link, motivating payment.

Student Performance Recording

- 1 **Record marks** — Enter scores for individual students per subject and form level, including a note or context (e.g., "CAT 1", "End of term exam").

- 2 **Performance history** — Each student has a performance log visible to their teacher and to themselves.
- 3 **Simple progress indicators** — A basic average score per subject is computed and displayed. No complex analytics in V1.



Teacher-Admin Functions

Elevated role

- 1 **Student onboarding management** — Review pending student registrations, approve or reject them, and send admission credentials.
- 2 **Payment confirmation** — Receive payment notification (M-Pesa reference or screenshot), manually mark the payment as confirmed, and toggle the student's access to e-library and/or e-classes.
- 3 **Student list & access view** — See all registered students, their payment status, and access flags in a simple table.

04 · STUDENT FEATURES

What Students Can Do

The student experience is designed around three states: **unpaid preview**, **paid access**, and **always-on historical data**. This creates a natural conversion funnel while protecting the value of existing work.

E-Library

- 1 **Browse & preview** — All students (paid and unpaid) can browse library content by subject, topic, and form level. Unpaid students see a content preview (first section/paragraph) with a payment prompt.
- 2 **Full access** — Paid and admin-enabled students access complete content including videos.

- 3 **PDF download** — Paid students can download any text/image content as a formatted PDF for offline study.
 - 4 **Previous academic data** — Students can always access their own performance records, marked assignments, and feedback regardless of payment status.
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Payments & Class Access

- 1 **View fees** — Prescribed fee amounts are displayed per term/package on the student's dashboard.
 - 2 **Pay externally** — Students pay via M-Pesa (or other method) and submit proof to the Teacher-Admin. No in-app payment gateway in V1.
 - 3 **Access unlock** — Once confirmed by Teacher-Admin, library and/or class access is toggled on immediately.
 - 4 **E-class preview (inactive)** — Upcoming classes show title, subject, teacher name, and date to unpaid students — but the Join button is hidden. A clear "Pay to join" prompt is shown.
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Login & Authentication

- 1 **Login with Admission Number + Password** — Students never need to remember an email. Their admission number is their username.
 - 2 **Default password = Date of birth** — Format DDMMYYYY. E.g., born 3 March 2009 → password is 03032009.
 - 3 **Forced password change on first login** — Students must set a new password before accessing the platform.
 - 4 **Password reset via parent phone** — OTP sent to registered parent phone number for recovery.
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Personal Tuition Requests

- 1 **Browse teachers** — Students can see a list of teachers on the platform, their subjects, and availability status.
- 2 **Submit tuition request** — Send a request to a specific teacher noting preferred subject, availability, and a short message.
- 3 **Teacher response** — Teacher accepts or declines; a Google Meet session can be manually scheduled from there. Full personal tuition scheduling is a V2 feature.

05 · ADDED VALUE FEATURES

Recommended Additions

The following features are recommended additions that are low-effort and high-impact for the V1 experience.

Inbuilt Academic Calendar

A shared calendar visible to both teachers and students showing:

- Scheduled e-classes (with subject, teacher, time)
- Assignment due dates
- National school holiday dates (pre-loaded for Kenyan academic calendar)
- Month and week views

Notification System

A lightweight in-app notification centre:

EVENT	WHO GETS NOTIFIED	CHANNEL
New class scheduled	Enrolled students	In-app
Assignment posted	Assigned students	In-app

EVENT	WHO GETS NOTIFIED	CHANNEL
Assignment marked	Student	In-app
Payment confirmed	Student	In-app
Registration approved	Student & parent	WhatsApp
Class reminder (1 hr prior)	Enrolled students	In-app
New e-library content	All students (paid)	In-app
Tuition request received	Teacher	In-app



Role Dashboards



Student Dashboard

Today's classes, recent library additions, assignment due dates, performance summary, and payment/access status.



Teacher Dashboard

Upcoming classes, pending submissions, recent student performance entries, and content published.



Admin Dashboard

Pending registrations, payment confirmations queue, total active students, and access status overview.

User Data & Privacy

The platform collects personally identifiable information (PII) for minors. The following safeguards are built into V1:



Parent consent is mandatory before a student account is created.



Student data is not displayed publicly. Teacher-Admins see only names, admission numbers, and access status.

- **No social features** — no student-to-student messaging or public profiles in V1.
- **Passwords are hashed** and never stored in plain text.
- **Data residency:** Preferred hosting within Africa (e.g., AWS af-south-1 or Azure South Africa) to comply with Kenya Data Protection Act 2019.

Suggested Technology Stack

Zafrika Tech Lab will use the following stack for a fast, maintainable V1 build:

FRONTEND React.js + TailwindCSS	BACKEND / API Django Rest Framework	DATABASE PostgreSQL
FILE / VIDEO STORAGE AWS S3 or Cloudinary	VIRTUAL CLASSES Google Meet API	AUTHENTICATION JWT + bcrypt
PDF GENERATION Puppeteer / WeasyPrint	HOSTING Digital Oceans	



Stack Flexibility

The final technology choices will be confirmed in the technical specification document. The above represents the recommended starting point based on cost, speed of development, and local ecosystem support.

07 · DELIVERY

Beta Roadmap

The following phased delivery plan ensures the platform is ready before the next major school holiday window.



PHASE 1 — WEEKS 1-2

Foundation & Onboarding

Database schema, authentication system, student and teacher registration forms, admission number generation, parent consent flow, and role-based access control.



PHASE 2 – WEEKS 3-4

Content & E-Library

Rich text editor, image upload, video upload and storage, e-library browse/search/filter, content preview for unpaid users, and PDF download for paid users.



PHASE 3 – WEEKS 5-6

Classes, Assignments & Payments

Google Meet class scheduling, student access gating, assignment creation and submission, marking and feedback, payment status management by Teacher-Admin.



PHASE 4 – WEEKS 7-8

Dashboards, Calendar & Notifications

Role-specific dashboards, academic calendar, in-app notification centre, SMS integration, performance recording, personal tuition request feature.



PHASE 5 – WEEK 9

QA, Testing & Beta Launch

Internal QA, pilot testing with a small cohort of teachers and students, bug fixes, performance checks, and deployment to production environment.

SIGN OFF —————

Approval & Agreement

This document represents the agreed scope for Hone Your Mark Version 1 (Beta). By signing below, both parties confirm that the features described are understood, agreed upon, and form the basis of the development engagement.



Note on Scope Changes

Any features not listed in this document requested during V1 development will be logged for V2 consideration. Additions to V1 scope require a written amendment signed by both parties.

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